SUBSCRIPTION MANAGEMENT · SALESFORCE TO AZURE

Low-level design

Agent 02 · Modelling · tables, rules, automations, security

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Figure 1. Order of execution: subscription line

Field lists, rules and automations are read from the org; each item shows its target design on Azure Database for PostgreSQL and the decision record behind it.

1. Tables

Account (`Account` → `account`)

22 rows. Sharing: ReadWrite.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Account ID	id	id	varchar(18)	yes	
Name	Account Name	string	name	varchar(255)	yes	
Type	Account Type	picklist	type	text		values: Prospect, Customer - Direct, Customer - Channel, Channel Partner / Reseller, Installation Partner, Technology Partner, Other
BillingCountry	Billing Country	string	billing_country	varchar(80)		
SLA__c	SLA	picklist	sla	text		values: Gold, Silver, Platinum, Bronze

Contact (`Contact` → `contact`)

20 rows. Sharing: ControlledByParent.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Contact ID	id	id	varchar(18)	yes	
Name	Full Name	string	name	varchar(121)		

Opportunity (`Opportunity` → `opportunity`)

31 rows. Sharing: ReadWrite.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Opportunity ID	id	id	varchar(18)	yes	
Name	Name	string	name	varchar(120)	yes	
StageName	Stage	picklist	stage_name	text	yes	values: Prospecting, Qualification, Needs Analysis, Value Proposition, Id. Decision Makers, Perception Analysis, Proposal/Price Quote, Negotiation/Review, Closed Won, Closed Lost
Amount	Amount	currency	amount	numeric(18,2)		
Loss_Reason__c	Loss Reason	picklist	loss_reason	text		values: Price, Timing, Competitor, No Budget, No Decision, Other

Product (`Product2` → `product2`)

17 rows. Sharing: ReadWrite.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Product ID	id	id	varchar(18)	yes	
Name	Product Name	string	name	varchar(255)	yes	

Subscription (`Subscription__c` → `subscription`)

107 rows. Sharing: ControlledByParent, inherited through Subscription__c.Account__c.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Record ID	id	id	varchar(18)	yes	
Name	Subscription Number	string	subscription_number	varchar(80)		
Account__c	Account	reference	account_id	varchar(18)	yes	→
Auto_Renew__c	Auto Renew	boolean	auto_renew	boolean		default false
Cancellation_Reason__c	Cancellation Reason	string	cancellation_reason	varchar(255)		
End_Date__c	End Date	date	end_date	date	yes	
Health_Score__c	Health Score	double	health_score	numeric(3,0)		
MRR__c	MRR	currency	monthly_recurring_revenue	numeric(16,2)		
Primary_Contact__c	Primary Contact	reference	primary_contact_id	varchar(18)		→
Source_Opportunity__c	Source Opportunity	reference	source_opportunity_id	varchar(18)		→
Start_Date__c	Start Date	date	start_date	date	yes	
Status__c	Status	picklist	status	text	yes	values: Draft, Active, Suspended, Cancelled, Renewed; default Draft
Term_Months__c	Term (Months)	double	term_months	numeric(4,0)		
Line_Count__c	Line Count	double	line_count	numeric(18,0)		
Annual_Value__c	Annual Value	currency	annual_value	numeric(18,2)		
Days_To_Renewal__c	Days To Renewal	double	days_to_renewal	numeric(18,0)		
Total_Line_Value__c	Total Line Value	currency	total_line_value	numeric(18,2)		

Subscription Line (`Subscription_Line__c` → `subscription_line`)

219 rows. Sharing: ControlledByParent, inherited through Subscription_Line__c.Subscription__c.

Salesforce field	Label	Salesforce type	Column	PostgreSQL type	Required	Notes
Id	Record ID	id	id	varchar(18)	yes	
Name	Line Number	string	line_number	varchar(80)		
Subscription__c	Subscription	reference	subscription_id	varchar(18)	yes	→
Product__c	Product	reference	product_id	varchar(18)		→
Quantity__c	Quantity	double	quantity	numeric(6,0)	yes	
Unit_Price__c	Unit Price	currency	unit_price	numeric(16,2)	yes	
Account_Name__c	Account Name	string	account_name	varchar(1300)		
Line_Total__c	Line Total	currency	line_total	numeric(18,2)		

2. Calculated fields

Field	Kind	Salesforce definition	Target design
Subscription_Line__c.Account_Name__c	formula	Subscription__r.Account__r.Name	view that joins the parent records ADR-004
Subscription_Line__c.Line_Total__c	formula	Quantity__c * Unit_Price__c	generated column, stored ADR-003
Subscription__c.Line_Count__c	rollup	count of undefined	column kept current by triggers, with a recompute job ADR-006
Subscription__c.Annual_Value__c	formula	MRR__c * 12	generated column, stored ADR-003
Subscription__c.Days_To_Renewal__c	formula	End_Date__c - TODAY()	computed when read (depends on today's date) ADR-005
Subscription__c.Total_Line_Value__c	rollup	sum of Subscription_Line__c.Line_Total__c	column kept current by triggers, with a recompute job ADR-006

3. Validation rules

Rule	Rejects when	Message	Target design
Opportunity: Closed_Won_Requires_Amount	AND(ISPICKVAL(StageName, "Closed Won"), OR(ISBLANK(Amount), Amount <= 0))	A won opportunity must have an Amount greater than zero.	CHECK constraint on the table ADR-007
Opportunity: Loss_Reason_Required_When_Lost	AND(ISPICKVAL(StageName, "Closed Lost"), ISBLANK(TEXT(Loss_Reason__c)))	A Loss Reason is required before an opportunity can be marked Closed Lost.	CHECK constraint on the table ADR-007
Subscription Line: Parent_Not_Cancelled	ISPICKVAL(Subscription__r.Status__c, "Cancelled")	You cannot add or change lines on a cancelled subscription.	BEFORE trigger guard (needs the previous row) ADR-008
Subscription Line: Quantity_Must_Be_Positive	Quantity__c <= 0	Quantity must be greater than zero.	CHECK constraint on the table ADR-007
Subscription: Active_Requires_MRR	AND(ISPICKVAL(Status__c, "Active"), OR(ISBLANK(MRR__c), MRR__c <= 0))	An active subscription must have an MRR greater than zero.	CHECK constraint on the table ADR-007
Subscription: Cancellation_Reason_Required	AND(ISPICKVAL(Status__c, "Cancelled"), ISBLANK(Cancellation_Reason__c))	Cancellation Reason is required when Status is Cancelled.	CHECK constraint on the table ADR-007
Subscription: End_Date_After_Start_Date	AND(NOT(ISBLANK(Start_Date__c)), NOT(ISBLANK(End_Date__c)), End_Date__c <= Start_Date__c)	End Date must be after Start Date.	CHECK constraint on the table ADR-007
Subscription: No_Reactivation_After_Cancel	AND(ISCHANGED(Status__c), ISPICKVAL(PRIORVALUE(Status__c), "Cancelled"), NOT(ISPICKVAL(Status__c, "Cancelled")))	A cancelled subscription cannot be reactivated. Create a new subscription instead.	BEFORE trigger guard (needs the previous row) ADR-008

4. Automations

Automation	Runs when	Does	Target design	Status
flow: Subscription At Risk Alert	after insert/update of Subscription	create Task	in-transaction hook after save	pending

Automation	Runs when	Does	Target design	Status
apexTrigger: OpportunityTrigger	after update of Opportunity	OpportunityService.createSubscriptionsForWonDeals	service hook in the same transaction	pending
apexTrigger: SubscriptionTrigger	before/before/after insert/update/update of Subscription	SubscriptionService.applyDefaults, SubscriptionService.handleStatusChange	service hook in the same transaction	partly translated
apexClass: SubscriptionRenewalBatch	on a schedule	writes Task	scheduled worker job	pending
duplicateRule: Account.Standard_Account_Duplicate_Rule	before insert/update of Account		rule engine hook in the service layer	pending
duplicateRule: Contact.Standard_Contact_Duplicate_Rule	before insert/update of Contact		rule engine hook in the service layer	pending

4.1 Order of execution: subscription

Shown in HLD section 6.3, Saving a subscription.

4.2 Order of execution: subscription line

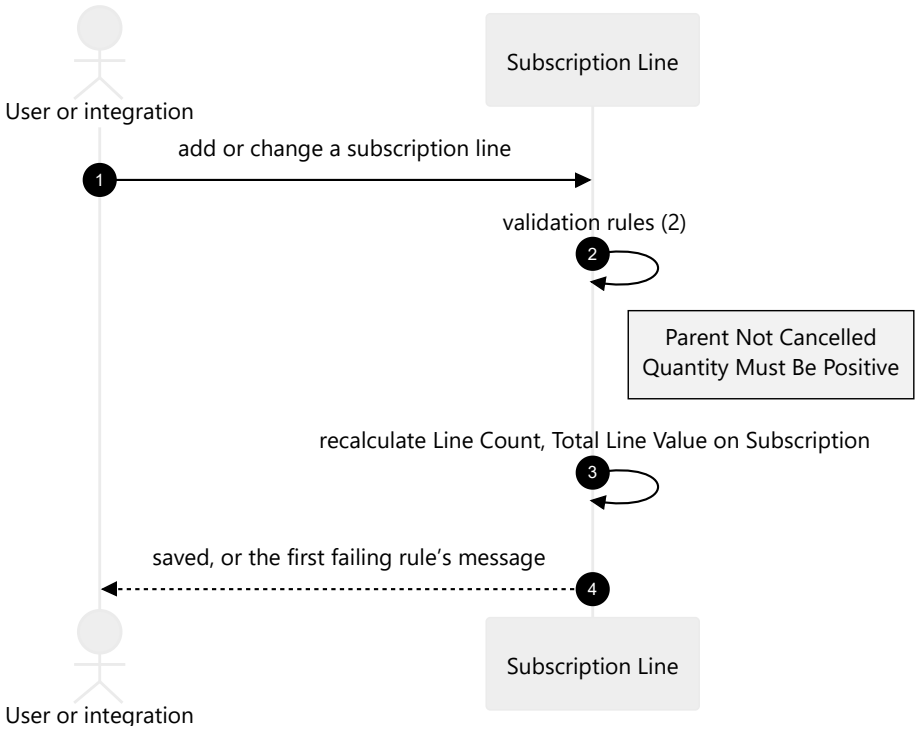


Figure 1. *Order of execution: subscription line*

4.3 Flow: Subscription At Risk Alert

Shown in HLD section 6.4, Subscription At Risk Alert.

5. Services

Class	Methods	Reads	Writes	Called by	Constants
OpportunityService	createSubscriptionsForWonDeals	Subscription	Subscription	OpportunityTrigger	DEFAULT_TERM_MONTHS = 12
SubscriptionService	applyDefaults, handleStatusChange, totalActiveMrr	Subscription	Task	SubscriptionTrigger	DEFAULT_HEALTH_SCORE = 70, AT_RISK_THRESHOLD = 50

6. Security

Role (from Salesforce)	Subscription	Subscription Line
Profile: Admin	read, create, edit, delete, viewAll, modifyAll	read, create, edit, delete, viewAll, modifyAll
Profile: Analytics Cloud Integration User	read, viewAll	read, viewAll
Permission set: Subscription_Management	read, create, edit, delete	read, create, edit, delete

How roles, column grants and row-level security fit together is shown in HLD section 8.2, Security across domains.

7. Data load

Order	Object	Rows	Depends on
1	Account	22	—
1	Product	17	—
2	Contact	20	Account
3	Opportunity	31	Account, Contact
4	Subscription	107	Account, Contact, Opportunity
5	Subscription Line	219	Subscription, Product

The load levels, and the migration waves this domain belongs to, are shown in HLD section 8.3.

8. Migrations

Order	Script	Contents
1	`000_runtime.sql`	runtime helpers (time zone, load mode)
2	`010_tables.sql`	tables, keys, constraints and generated columns
3	`020_views.sql`	views for cross-object and date-dependent fields
4	`030_rollups.sql`	roll-up totals and their recompute function
5	`040_guards.sql`	rule guards that need the previous row or a parent
6	`050_translations.sql`	translated business logic
7	`060_security.sql`	roles, column grants and row-level security

Scripts are numbered and applied in order by the release pipeline; each is idempotent, so any environment can be rebuilt.

9. Patterns used

Pattern	Where	Why
Generated column	row-local formulas	Always correct without application code

Pattern	Where	Why
Materialised aggregate with recompute backstop	roll-up summaries	Fast reads, self-healing after loads
Guard trigger	rules needing the previous row or a parent	Same rejection point and message as Salesforce
Role-based access + row-level security	`060_security.sql`	Access enforced in the database for every client
Idempotent, dependency-ordered load	data migration	Repeatable loads; triggers deferred, totals recomputed once
Golden-master testing (parity)	parity tests	Salesforce's own values are the expected results

10. Verification cases

Case	Comes from
apex.appliesDefaultHealthScoreAndEndDate	apex-test:SubscriptionServiceTest#appliesDefaultHealthScoreAndEndDate
apex.blocksReactivationOfCancelledSubscription	apex-test:SubscriptionServiceTest#blocksReactivationOfCancelledSubscription
apex.rollsUpLineTotals	apex-test:SubscriptionServiceTest#rollsUpLineTotals
apex.createsSubscriptionWhenOpportunityIsWon	apex-test:SubscriptionServiceTest#createsSubscriptionWhenOpportunityIsWon
apex.renewalBatchRaisesTasks	apex-test:SubscriptionServiceTest#renewalBatchRaisesTasks
rule.Active_Requires_MRR.blank	rule
rule.Active_Requires_MRR.zero	rule
rule.Active_Requires_MRR.draftWithoutMrrAllowed	rule
rule.Cancellation_Reason_Required	rule
rule.End_Date_After_Start_Date	rule
rule.No_Reactivation_After_Cancel.toSuspended	rule
rule.Parent_Not_Cancelled	rule
rule.Quantity_Must_Be_Positive	rule
platform.restrictedPicklist	rule
platform.masterDetailCascade	rule
apex.applyDefaults.termFromDates	rule
security.permsetEditsAllowedFields	rule
security.readOnlyFieldsNotWritable	rule
security.readOnlyProfileCannotEdit	rule

11. Traceability

Each Salesforce element, its blueprint ID, the design decision applied, the generated artifact, and the test that proves it.

Salesforce element	Blueprint ID	Design	Artifact	Verified by
formula: `Subscription_Line__c.Account_Name__c`	Subscription_Line__c.Account_Name__c	view that joins the parent records ADR-004	`schema/sql/020_views.sql`	derived-value comparison (all rows)
formula: `Subscription_Line__c.Line_Total__c`	Subscription_Line__c.Line_Total__c	generated column, stored ADR-003	`schema/sql/010_tables.sql`, `schema/sql/030_rollups.sql`	derived-value comparison (all rows)

Salesforce element	Blueprint ID	Design	Artifact	Verified by
rollup: `Subscription__c.Line_Count__c`	Subscription__c.Line_Count__c	column kept current by triggers, with a recompute job ADR-006	`schema/sql/010_tables.sql`, `schema/sql/030_rollups.sql`	derived-value comparison (all rows)
formula: `Subscription__c.Annual_Value__c`	Subscription__c.Annual_Value__c	generated column, stored ADR-003	`schema/sql/010_tables.sql`	derived-value comparison (all rows)
formula: `Subscription__c.Days_To_Renewal__c`	Subscription__c.Days_To_Renewal__c	computed when read (depends on today's date) ADR-005	`schema/sql/020_views.sql`	derived-value comparison (all rows)
rollup: `Subscription__c.Total_Line_Value__c`	Subscription__c.Total_Line_Value__c	column kept current by triggers, with a recompute job ADR-006	`schema/sql/010_tables.sql`, `schema/sql/030_rollups.sql`	derived-value comparison (all rows)
validation rule: `Opportunity.Closed_Won_Requires_Amount`	Opportunity.Closed_Won_Requires_Amount	CHECK constraint on the table ADR-007	—	—
validation rule: `Opportunity.Loss_Reason_Required_When_Lost`	Opportunity.Loss_Reason_Required_When_Lost	CHECK constraint on the table ADR-007	—	—
validation rule: `Subscription_Line__c.Parent_Not_Cancelled`	Subscription_Line__c.Parent_Not_Cancelled	BEFORE trigger guard (needs the previous row) ADR-008	`schema/sql/010_tables.sql`, `logic/sql/040_guards.sql`	`rule.Parent_Not_Cancelled`
validation rule: `Subscription_Line__c.Quantity_Must_Be_Positive`	Subscription_Line__c.Quantity_Must_Be_Positive	CHECK constraint on the table ADR-007	`schema/sql/010_tables.sql`	`rule.Quantity_Must_Be_Positive`
validation rule: `Subscription__c.Active_Requires_MRR`	Subscription__c.Active_Requires_MRR	CHECK constraint on the table ADR-007	`schema/sql/010_tables.sql`	`rule.Active_Requires_MRR.blank`, `rule.Active_Requires_MRR.zero`, `rule.Active_Requires_MRR.draftWithoutMrrAllowed`
validation rule: `Subscription__c.Cancellation_Reason_Required`	Subscription__c.Cancellation_Reason_Required	CHECK constraint on the table ADR-007	`schema/sql/010_tables.sql`	`rule.Cancellation_Reason_Required`
validation rule: `Subscription__c.End_Date_After_Start_Date`	Subscription__c.End_Date_After_Start_Date	CHECK constraint on the table ADR-007	`schema/sql/010_tables.sql`	`rule.End_Date_After_Start_Date`
validation rule: `Subscription__c.No_Reactivation_After_Cancel`	Subscription__c.No_Reactivation_After_Cancel	BEFORE trigger guard (needs the previous row) ADR-008	`schema/sql/010_tables.sql`, `logic/sql/040_guards.sql`	`rule.No_Reactivation_After_Cancel.toSuspended`
flow: `Subscription_At_Risk_Alert`	flow:Subscription_At_Risk_Alert	in-transaction hook after save	pending	—
apexTrigger: `OpportunityTrigger`	apex-trigger:OpportunityTrigger	service hook in the same transaction ADR-010	pending	—

Salesforce element	Blueprint ID	Design	Artifact	Verified by
apexTrigger: `SubscriptionTrigger`	apex-trigger:SubscriptionTrigger	service hook in the same transaction ADR-010	`logic/sql/050_translations.sql`	`apex.applyDefaults.termFromDates`
apexClass: `SubscriptionRenewalBatch`	apex-job:SubscriptionRenewalBatch	scheduled worker job	pending	—
duplicateRule: `Account.Standard_Account_Duplicate_Rule`	duplicateRule:Account.Standard_Account_Duplicate_Rule	rule engine hook in the service layer	pending	—
duplicateRule: `Contact.Standard_Contact_Duplicate_Rule`	duplicateRule:Contact.Standard_Contact_Duplicate_Rule	rule engine hook in the service layer	pending	—
profile: `Admin`	profile:Admin	role, column grants, RLS ADR-009	`security/sql/060_security.sql`	`security.permsetEditsAllowedFields`, `security.readOnlyFieldsNotWritable`, `security.readOnlyProfileCannotEdit`
profile: `Analytics Cloud Integration User`	profile:Analytics Cloud Integration User	role, column grants, RLS ADR-009	—	`security.permsetEditsAllowedFields`, `security.readOnlyFieldsNotWritable`, `security.readOnlyProfileCannotEdit`
permissionSet: `Subscription_Management`	permissionSet:Subscription_Management	role, column grants, RLS ADR-009	`security/sql/060_security.sql`	`security.permsetEditsAllowedFields`, `security.readOnlyFieldsNotWritable`, `security.readOnlyProfileCannotEdit`